

Head CT Angiography (CTA)

ACG: A-0484 (AC)

[Link to Codes](#)

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Clinical Indications

- Head CT angiography (CTA) may be indicated for **1 or more** of the following(1)(2)(3)(4)(5):
 - ☐ Aneurysm or arteriovenous malformation, known or suspected, as indicated by **1 or more** of the following(9)(10)(11):[N](#)
 - Follow-up of known aneurysm or arteriovenous malformation(33)
 - Initial evaluation of suspected aneurysm or arteriovenous malformation (eg, acute severe headache or diagnosis of intracerebral or subarachnoid hemorrhage)(34)(35)(36)(37)(38)
 - Intracerebral hemorrhage, known, and hematoma expansion suspected(29)(36)(39)
 - Postoperative evaluation of intracranial aneurysm after surgical therapy(40)
 - Preoperative or preprocedural planning prior to endovascular therapy (eg, coiling, embolism)(41)
 - Subarachnoid hemorrhage, and evaluation of possible intracranial vasospasm needed(31)(42)(43)
 - Intracranial arterial dissection, suspected(38)(44)(45)(46)(47)[N](#)
 - Intracranial arterial occlusive disease, suspected, when imaging results will impact patient management(38)(50)(51)(52)(53)(54)[N](#)
 - Intracranial vasculitis, suspected(11)(75)(76)(77)[N](#)
 - Intracranial venous occlusive disease, suspected(36)(81)(82)(83)(84)[N](#)
 - Preoperative planning for delineation of vascular supply of vascular neoplasm(1)(87)[N](#)
 - Preprocedural planning for carotid endarterectomy or percutaneous intervention for acute stroke, when performed in conjunction with neck CTA(56)(88)(89)(90)[N](#)
 - ☐ Screening for cerebral artery aneurysm in patient with **1 or more** of the following(67)(92)(93):[N](#)
 - Autosomal dominant polycystic kidney disease(95)(96)(97)
 - Two or more family members with intracranial aneurysm or subarachnoid hemorrhage
 - Tinnitus, pulsatile(98)(99)[N](#)
 - Traumatic injury to cervicocerebral vessels, suspected(100)(101)(102)(103)(104)(105)[N](#)
 - Vascular malformation of brain or skull base, for confirmation of diagnosis or preoperative planning(45)(113)(114)(115)[N](#)
 - ☐ Repeat evaluation of specific area or structure with same imaging modality, and **ALL** of the following:
 - Clinical need for repeat imaging, as indicated by **1 or more** of the following:
 - Change in clinical status (eg, worsening symptoms or new associated symptoms), and findings may impact treatment
 - Need for interval reassessment that may impact treatment plan
 - Need for re-imaging either prior to or after performance of invasive procedure
 - Prior imaging results of specific area or structure with same imaging modality documented and available for comparison

Evidence Summary

Background

CTA consists of CT scanning after intravenous injection of contrast material to characterize vascular anatomy and pathology. Unlike standard invasive x-ray and digital subtraction angiography, it is noninvasive and rapidly provides information about extravascular and perivascular anatomy. CTA multislice helical scanning with 3-dimensional reconstruction software can provide detailed, visually familiar views of complex anatomy. However, the procedure involves exposure to radiation and contrast material, with risk of allergic reaction and nephrotoxicity.(1)(2)(6)(7)(8) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 95 published peer reviewed articles, 11 specialty society or other evidence-based guidelines, 1 Cochrane systematic review, and 1 book section.

For cerebral aneurysm or arteriovenous malformation, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Catheter angiography remains the gold standard for diagnosis and formal delineation of cerebral aneurysms or arteriovenous malformations; however, at many institutions, diagnostic CTA of the head has largely replaced invasive angiography.(10) (11)(12) **(EG 2)** A meta-analysis of 8 studies with 982 patients found that subtraction CTA had sensitivity of 99% and specificity of 94% for detecting cerebral aneurysms when compared with digital subtraction angiography.(13) **(EG 1)** Cohort studies of patients with intracranial aneurysms who underwent both CTA and catheter angiography revealed that CTA had sensitivity of 90.2% to 100% and specificity of 93.4% to 100% on a per-aneurysm basis, with slightly decreased sensitivity for aneurysms smaller than 3 mm in diameter. (14)(15)(16) **(EG 2)** For diagnosing cerebral aneurysms, a meta-analysis of 10 studies comparing CTA with MRA found that CTA had a pooled sensitivity of 84% and specificity of 85% and MRA had a pooled sensitivity of 80% and specificity of 87%; the authors concluded that both CTA and MRA have high accuracy for diagnosing cerebral aneurysms, particularly when using multidetector CT technology. (17) **(EG 1)** A retrospective review of 1366 patients found that CTA had a sensitivity of 97.1% to 97.4% and a specificity of 98.5% to 99.1% for diagnosing cerebral aneurysms 5 mm or smaller, compared with digital subtraction angiography.(18) **(EG 2)** For patients who presented with subarachnoid hemorrhage, a systematic review and meta-analysis of 50 studies showed that CTA had a pooled sensitivity of 98% and specificity of 100% for diagnosis of cerebral aneurysms.(19) **(EG 1)** A specialty society guideline recommends noncontrast head CT scan followed by either lumbar puncture or CTA for evaluating patients with suspected subarachnoid hemorrhage. (20) **(EG 2)** CTA is an acceptable alternative initial technique for cerebral aneurysm or arteriovenous malformation visualization, and 3-dimensional or 4-dimensional reconstruction of images can assist in preoperative planning.(21)(22)(23)(24) **(EG 2)** Review articles note that head CTA is a commonly used noninvasive imaging technique for ruling out or confirming vascular abnormalities as a secondary cause of intracerebral or subarachnoid hemorrhage.(6)(25)(26) **(EG 2)** A meta-analysis of 5 studies comparing the diagnostic accuracy of CTA with digital subtraction angiography in patients with acute spontaneous intracerebral hemorrhage concluded that CTA with venography was accurate enough to substitute for digital subtraction angiography in the vascular evaluation of acute spontaneous intracerebral hemorrhage.(27) **(EG 2)** A meta-analysis and review articles note that CTA, via what is known as the CTA "spot sign," may be useful for identifying expansion of the hematoma caused by intracerebral hemorrhage.(28)(29)(30) **(EG 1)** A meta-analysis of 6 studies showed that CTA had a pooled sensitivity of 79.6% and specificity of 93.1% for evaluation of vasospasm following subarachnoid hemorrhage when compared with digital subtraction angiography as the reference standard.(31) **(EG 1)** Appropriate-use criteria state that angiography is appropriate in children as a follow-up test if subarachnoid hemorrhage or nontraumatic intraparenchymal hemorrhage is seen on CT scan or MRI.(32) **(EG 2)**

For intracranial arterial dissection, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A systematic review noted that initial evaluation of carotid and vertebral artery dissection is usually made by MRI in combination with CTA or MRA; the authors concluded that test characteristics among the 3 imaging modalities were relatively similar for the diagnosis of carotid and vertebral artery dissection.(44) **(EG 2)** A retrospective study of 40 patients with 43 spontaneous cervical artery dissections and 26 control patients without dissections found that CTA was 100% sensitive and specific as compared with a diagnostic standard of MRA.(48) **(EG 2)** A review article notes that dissected aneurysms of the petrous and cavernous segments of the internal carotid artery at the skull base are best diagnosed with digital subtraction angiography, CTA, or MRA.(45) **(EG 2)** Appropriate-use criteria state that angiography may be appropriate for evaluation of known or suspected cranial or cervical arterial dissection in children and adults.(32) (38) **(EG 2)** Appropriate-use criteria state that head CTA is appropriate for evaluation of patients with symptoms of vertebrobasilar insufficiency (ie, chronic recurrent vertigo, brainstem neurologic deficits) when vertebral artery dissection is suspected.(49) **(EG 2)**

For intracranial arterial occlusive disease, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** MRA or CTA are appropriate initial imaging techniques for detection of significant stenoses, and reviews have reported sensitivity and specificity of close to 100%.(55)(56)(57)(58) **(EG 2)** An open-label randomized trial of 502 patients with acute ischemic stroke presenting 6 to 24 hours from symptom onset or last seen well (all of whom had anterior circulation large vessel occlusion and collateral flow on CTA) compared treatment with best medical care with or without endovascular therapy and found, at 90-day follow-up, that patients treated with endovascular therapy had lower modified Rankin Scale scores compared with patients not treated with endovascular therapy; no difference was seen in all-cause mortality between groups, and patients treated with endovascular therapy had higher rates of intracranial hemorrhage compared with patients not treated with endovascular therapy.(59) **(EG 1)** Analysis of data from a randomized trial of 656 patients with acute ischemic stroke with large vessel occlusion revealed that the status of collateral vessels on baseline CTA was a determinant of final clinical outcome and could be used to select patients for endovascular intervention. (60) **(EG 1)** The use of CTA did not delay treatment when compared with noncontrast CT scan alone.(61) **(EG 1)** A prospective cohort

study involving 84 patients with acute ischemic stroke concluded that patients with good or intermediate collateral blood flow on CTA derived benefit from intra-arterial therapy, whereas patients with poor collaterals did not, indicating that the use of head CTA for assessment of collateral status in acute ischemic stroke can impact treatment and management.(62) **(EG 2)** Analysis of a multicenter stroke registry (including 1412 patients with an anterior circulation large vessel occlusion) found, at 90-day follow-up, that better collateral blood flow on baseline CTA was associated with improved functional outcomes measured by modified Rankin Scale score.(63) **(EG 2)** Analysis from a multicenter randomized trial of endovascular treatment for acute ischemic stroke (500 patients) and from a multicenter stroke registry (including 1488 patients with anterior circulation large vessel occlusion) found that intracranial internal carotid artery occlusions with terminus occlusions were associated with poor collateral grades as measured by CTA. However, findings were limited by potential for selection bias and interobserver variability in CTA readings; further well-designed long-term studies were recommended.(64) **(EG 2)** Analysis of a multicenter stroke registry (including 1265 patients) comparing CTA with MRI for selecting candidates for endovascular therapy found, at 3 months post intervention, no difference in excellent or good (defined as modified Rankin Scale score of 0 to 2) functional outcomes between the groups.(65) **(EG 2)** A prospective study of 510 patients with transient ischemic attack and minor stroke found that the use of CTA for early assessment of the intracranial and extracranial vasculature enabled prediction of recurrent stroke and clinical outcome.(66) **(EG 2)** Review articles note that CTA can quickly and accurately localize the site of occlusion in hyperacute ischemic stroke.(3)(67)(68)(69) **(EG 2)** Other review articles state that head and neck CTA should be performed to evaluate patients with suspected acute ischemic stroke to identify a potential documented occlusion of the distal internal carotid or the proximal middle cerebral artery to determine if intra-arterial thrombectomy is a treatment option.(70)(71) **(EG 2)** Practice guidelines recommend that CTA or MRA be available at primary stroke centers for admitted stroke patients who may benefit from such testing when needed to guide or determine treatment decisions; CTA or MRA is recommended as part of the initial evaluation of stroke patients who may be candidates for endovascular intervention.(38)(72)(73)(74) **(EG 2)**

For intracranial vasculitis, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A review article notes that CTA may be used for evaluation but carries the attendant risks of radiation and contrast exposure.(76) **(EG 2)** Other review articles note that CTA may be useful for the evaluation of cranial giant cell arteritis.(78)(79)(80) **(EG 2)** A specialty society guideline states that CTA is appropriate for evaluation of intracranial vasculitis and collagen vascular diseases.(1) **(EG 2)** Another specialty society guideline states that CTA can detect intracranial vessel luminal features of CNS vasculitis, including multifocal vessel wall narrowing and dilatation; the sensitivity of CTA is limited by resolution, and findings can overlap with nonvasculitic cerebral vascular diseases.(11) **(EG 2)** Appropriate-use criteria state that angiography may be appropriate in children with symptoms suggestive of stroke and known or suspected intracranial vasculitis.(32) **(EG 2)**

For intracranial venous disease (eg, venous sinus thrombosis), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A specialty society guideline states that CTA and MRA are appropriate first-line imaging options for evaluation of intracranial venous occlusive disease.(83) **(EG 2)** However, catheter angiography remains the definitive diagnostic reference standard and is appropriate whenever results from noninvasive imaging are indeterminate, or when endovascular intervention or thrombolysis is being considered.(81)(82)(85) **(EG 2)** Review articles note that CTA can be used to detect cerebral venous thrombosis through visualization of venous structures and demonstration of intraluminal defects.(36)(86) **(EG 2)**

For preoperative planning for delineation of vascular supply of vascular neoplasm, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline states that CTA is appropriate for the evaluation of head and neck tumors of vascular origin, particularly with a rich vascular supply or invading vascular structures.(1) **(EG 2)**

For preprocedural planning for vascular interventions, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline states that CTA is appropriate for preoperative guidance of percutaneous and surgical vascular interventions.(1) **(EG 2)** Another specialty society guideline notes that when intervention is planned for significant carotid stenosis detected by duplex sonography, CTA, MRA, or digital subtraction angiography of the head and neck can be useful to evaluate stenosis severity and to identify intracranial or intrathoracic vascular lesions that are inadequately assessed by duplex sonography.(91) **(EG 2)**

For screening for cerebral aneurysms, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A meta-analysis of 15 studies with 1490 patients with autosomal dominant polycystic kidney disease who had brain imaging with either MRA, CTA, or digital subtraction angiography found that 10% had intracranial aneurysms; a family history of intracranial aneurysm or hemorrhagic stroke significantly increased the risk of aneurysm occurrence.(94) **(EG 1)** A specialty society guideline recommends brain imaging with either MRA or CTA to screen for unruptured aneurysms in patients with either a history of autosomal dominant polycystic kidney disease (especially those with a family history of intracranial aneurysms) or 2 or more family members with intracranial aneurysms or subarachnoid hemorrhages.(92) **(EG 2)**

For pulsatile tinnitus, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** For patients with symptoms of pulsatile tinnitus, specialty society guidelines support head and neck imaging, including head CTA, to evaluate for possible carotid vascular lesions (eg, atherosclerosis, dissection, fibromuscular dysplasia) or intracranial vascular etiologies (eg, arteriovenous malformation, dural arteriovenous fistula, vascular tumor).(98)(99) **(EG 2)**

For traumatic injury to cervicocerebral vessels, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review of 21 studies recommended CTA as a screening imaging modality to evaluate patients for blunt cerebrovascular injury following craniomaxillofacial trauma.(106) **(EG 1)** A retrospective study of 45 consecutive patients with penetrating traumatic brain injuries found that CTA had sensitivity of 72.7% and specificity of 93.5% for detecting intracranial arterial injuries when compared with digital subtraction angiography; however, 100% of 7 traumatic intracranial aneurysms were correctly identified by CTA. The authors concluded that CTA was useful in finding traumatic intracranial aneurysms and could supersede digital subtraction angiography as the initial imaging modality of choice for evaluating patients with penetrating head injuries, but a negative test result should not preclude follow-up with conventional angiography.(107) **(EG 2)** A retrospective analysis of 3747 patients who underwent CTA for severe polytrauma found that 180 (4.8%) had sustained blunt cerebrovascular injuries; 128 patients had a follow-up CTA within 6 months, of whom 44% were improved and 51% were unchanged.(108) **(EG 2)** Specialty society guidelines and review articles state that CTA is an imaging modality of choice for the evaluation of cervicocerebral vascular trauma, including dissections, occlusions, or pseudoaneurysms.(1)(105)(109)(110)(111)(112) **(EG 2)**

For vascular malformations of the brain or skull base, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review of 8 studies that compared CTA with catheter-based digital subtraction angiography for the detection of intracranial vascular malformations following intracerebral hemorrhage concluded that CTA had a pooled sensitivity of 95% and specificity of 99%; however, the authors noted that several of the included studies had methodological shortcomings, such as retrospective design and verification bias, that may have increased apparent test accuracy.(116) **(EG 1)** A review article notes that CTA can be useful for the evaluation of aberrant carotid arteries and congenital anomalies of the cerebral vessels.(114) **(EG 2)** A specialty society guideline states that CTA is appropriate for the evaluation of congenital vascular anomalies and vascular anatomic variants.(1) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(117); 42 CFR 422.101(118)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(119); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(120); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(121)

Medicare Coverage Determinations: Medicare Coverage Database(122)

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